Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

Introduction to Bayesian Analysis (MTH422)

Quiz 4, Date: April 11, 2025, Friday

Time: 8:00–8:40 AM (8:00–8:55 AM for DAP students) Spring 2025 Max point: 10

The answers need to be explicit. Writing some philosophical answer and claiming that it is correct will lead to no point. Calculation mistakes will be penalized by 50% of the allotted marks.

1. Suppose the response variable Y_i is the binary indicator that child i tested positive for malaria, and we have covariate information $X_{ij}, j = 1, \ldots, p$ for $i = 1, \ldots, n$. We want to fit the logistic regression model $\text{logit}[\text{Prob}(Y_i = 1)] = \sum_{j=1}^{p} \beta_j X_{ij}$ and choose the priors $\beta_j \overset{\text{IID}}{\sim} \text{Normal}(0, \sigma_0^2)$. Calculate the MAP estimator of $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)'$. If no analytical solution is available, write the equation you need to solve to obtain the MAP estimator. (2 points)

2. Consider a two-way random effects model $Y_{i,j,k} = \mu + \alpha_i + \beta_j + \varepsilon_{i,j,k}$, where $\alpha_i \overset{IID}{\sim} \text{Normal}(0, \sigma_\alpha^2)$, $\beta_j \overset{IID}{\sim} \text{Normal}(0, \sigma_\beta^2)$, and $\varepsilon_{i,j,k} \overset{IID}{\sim} \text{Normal}(0, \sigma_\varepsilon^2)$. Find the correlation values $\text{Cor}(Y_{1,1,1}, Y_{2,1,1})$ and $\text{Cor}(Y_{1,2,3}, Y_{2,1,3})$. (1+1=2 points)

3. The following is a `R` snippet. Find the unconditional joint distribution of $\boldsymbol{X}$. (2 points)

```
mu <- rnorm(n = 2, mean = 1, sd = 1)
X <- rnorm(n = 4, mean = mu, sd = 1)
```

4. A semiparametric Bayesian regression model is $Y_i | f(\cdot) \sim \text{Normal}(f(X_i), \sigma^2)$, independently across the $i = 1, \ldots, n$ observations. Here, σ^2 is known. Let us assume that $f(x) = \sum_{\ell=1}^{L} \beta_\ell B_\ell(x), x \in [a, b]$. Here, $B_\ell(\cdot)$ are known cubic B-splines. Let $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_L)'$. Find the Jeffreys' prior for $\boldsymbol{\beta}$. (2 points)

5. We want to compare $\mathcal{M}_1 : Y \sim \text{Normal}(0, \sigma^2)$ and $\mathcal{M}_2 : Y \sim \text{Normal}(\mu, \sigma^2)$. Here, σ^2 is known and we choose the prior $\mu \sim \text{Normal}(\mu_0, \sigma_0^2)$. Find the Bayes factor $BF_{2,1}$ in terms of μ_0, σ_0^2, σ^2, and Y. (2 points)